Running the MLIP tutorials on Google Colab

Student guide for the three notebooks of `TP-MLIP`

Machine-learned interatomic potentials — hands-on session

The three tutorials on machine-learned interatomic potentials (MLIPs) run on **Google Colab**. All you need is a web browser and a Google account. Nothing has to be installed on your computer: Colab gives you a Linux machine with 2 CPUs, and the first cell of each notebook installs everything else on it.

Everything starts from the course page, which you can open with the QR code:

https://github.com/zacway/TP-MLIP

1 The three tutorials

Do them in order. Tutorials 1 and 2 each come with lesson notes that explain the theory and discuss the results: read them alongside the notebook. They are linked from the course page.

Tutorial	Notebook	Computing time
1. Zundel ion, $H_5O_2^+$	`zundel.ipynb`	∼10 min
2. Protonated water cluster, $(H_2O)_6H^+$	`wclusters.ipynb`	∼20 min
3. Published models for aqueous systems	`wsystems.ipynb`	∼20 min

The computing time is the total time spent waiting for cells to finish, setup included. Most of it goes into training the neural networks and running molecular dynamics (MD). Reading, thinking and answering the questions come on top of it.

2 Step by step

1. **Open the tutorial in Colab.** On the course page, click on the **Open in Colab** button of the tutorial. The notebook opens in Colab; sign in with your Google account if asked.

2. **Run the setup cell.** Click on the first code cell and press **Shift+Enter**. Colab warns that the notebook was not authored by Google: click **Run anyway**. The cell downloads the course files, installs n2p2 (to train the neural networks) and CP2K (to run MD), and moves into the tutorial's folder. It takes 1 to 2 minutes the first time, and prints `Setup complete.` when it is done.

If the setup cell stops with "Could not download TP-MLIP.zip"

Download `TP-MLIP.zip` yourself from the **Releases** section of the course page (right-hand side). In Colab, click on the folder icon in the left sidebar to open the **Files** panel, drag and drop the zip file into it, wait until the upload is finished, then run the setup cell again.

3. **Work through the notebook from top to bottom.** Run the cells one by one with **Shift+Enter**, reading the text as you go. The questions are written in blue. Each cell uses the results of the cells above it, so do not skip any.
   - MD cells call `run_cp2k(...)`. While the simulation runs, they print its latest step every 15 seconds, then `Done in X min.` Wait for that line before moving on.
   - Training cells take a few minutes as well. The active-learning cell prints a summary after each of its 15 iterations; again, wait until the cell has finished.
4. **Tutorials 2 and 3.** Open each notebook the same way (step 1) and run its setup cell first. The tutorials do not depend on each other: each one trains or downloads its own models.

3 Good to know

Keep the default CPU runtime. Do not switch to a GPU in **Runtime** > **Change runtime type**: n2p2 and CP2K only use the CPU.

Looking at and editing files. The tutorial files are in `/content/TP-MLIP`, which you can browse in the **Files** panel. Double-click on a text file, such as a CP2K input (`*.inp`), to open it in an editor next to the notebook, and press **Ctrl+S** to save your changes.

Colab's files are temporary. After some time without activity (typically around 90 minutes), when you close the tab for long, or after **Runtime** > **Disconnect and delete runtime**, Colab gives you a fresh machine and everything computed so far is lost: trained models, trajectories, installed programs. Your notebook and its displayed outputs remain. To carry on, click on the cell where you stopped and use **Runtime** > **Run before**: it runs the setup cell and all the other cells above it again.

Saving your work. The notebook opened from the course page is not saved anywhere. **File** > **Save a copy in Drive** keeps a copy of the notebook with your notes and outputs (plots, scores), but not the files the notebook produced. To keep one of those, use the ⋮ menu next to it in the **Files** panel and choose **Download**.

Training again with other settings. The notebooks invite you to try other network architectures. Before training again, delete the previous results, or they will interfere with the new ones. Run this in a new code cell (**+ Code**) before re-running the training cells:

```
!rm -rf {base_dir}/qbc/iteration-* {base_dir}/qbc/results.shelf*
!rm -rf {base_dir}/train/final_training
```

and, for the single model at the start of tutorial 1:

```
!rm -rf {base_dir}/single/training
```

Changing the number of committee members

The CP2K input files list the committee members one by one, in one `&MODEL` block each (8 blocks). If you change `n_members`, edit the `&MODEL` blocks of the inputs used afterwards so that they match: `nnp_md/zundel.inp` and `cluster/cluster.inp` in tutorial 1; `nnp_md/cluster.inp`, `big_cluster/big_cluster.inp` and `bulk/bulk.inp` in tutorial 2.

If something goes wrong. Use **Runtime** > **Restart session**, which keeps the files, then use **Runtime** > **Run before** on the cell you were at. If it is still broken, use **Runtime** > **Disconnect and delete runtime** and start again from the setup cell.